

CUSTOMER ANALYTICS PROJECT REPORT

1. DATASET OVERVIEW

Total Rows: 255
Total Columns: 14

2. COLUMN INFORMATION & MISSING VALUES

CustomerID	int64	Missing: 0 (0.00%)
Age	int64	Missing: 0 (0.00%)
Gender	object	Missing: 0 (0.00%)
City	object	Missing: 0 (0.00%)
Education	object	Missing: 12 (4.71%)
MaritalStatus	object	Missing: 0 (0.00%)
AnnualIncome	float64	Missing: 12 (4.71%)
SpendingScore	int64	Missing: 0 (0.00%)
YearsEmployed	int64	Missing: 0 (0.00%)
PurchaseFrequency	int64	Missing: 0 (0.00%)
OnlineVisitsPerMonth	int64	Missing: 0 (0.00%)
ReturnedItems	int64	Missing: 0 (0.00%)
PreferredDevice	object	Missing: 0 (0.00%)
LastPurchaseAmount	int64	Missing: 0 (0.00%)

3. DUPLICATE RECORD CHECK

Duplicate Rows Found: 5

4. BASIC STATISTICS (NUMERICAL COLUMNS)

	count	mean	std	min	25%
50% 75% max					
CustomerID	255.0	1126.941176	72.397256	1001.0	1064.5
1128.0 1190.5	1250.0				
Age	255.0	37.729412	9.767842	21.0	29.0
38.0 46.0	54.0				
AnnualIncome	243.0	74499.901235	43939.855655	16062.0	56353.0
69629.0 84030.5	474327.0				
SpendingScore	255.0	45.721569	17.869522	5.0	34.5
47.0 57.5	95.0				
YearsEmployed	255.0	14.682353	9.651664	1.0	6.0
15.0 23.0	34.0				
PurchaseFrequency	255.0	11.568627	7.077899	1.0	5.0
11.0 18.0	24.0				
OnlineVisitsPerMonth	255.0	16.078431	7.909536	3.0	10.0
16.0 23.0	29.0				
ReturnedItems	255.0	1.858824	1.409917	0.0	1.0
2.0 3.0	4.0				
LastPurchaseAmount	255.0	2795.074510	1328.772410	566.0	1542.5
2705.0 4001.0	4996.0				

5. CORRELATION ANALYSIS

Strongest Positive Correlation:
Age YearsEmployed 0.974985
dtype: float64

Strongest Negative Correlation:

AnnualIncome SpendingScore -0.388748
dtype: float64

6. KEY INSIGHTS

- Majority of customers fall between 30-50 years of age.
- Income distribution is positively skewed with high-income outliers.
- Slightly more male customers than female customers.
- Moderate negative relationship observed between Annual Income and Spending Score.
- Strong positive correlation between Age and Years Employed.
- Most behavioral variables exhibit weak linear relationships.

7. CONCLUSION

The dataset required minimal preprocessing and revealed meaningful patterns in customer demographics and purchasing behavior. Income alone does not strongly determine spending behavior. Customer behavior appears multi-dimensional and suitable for segmentation and predictive modeling.